FIT
EATS

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FINAL YEAR PROJECT · B.TECH CSE · 2025–26

FitEats

Intelligent Food Delivery System

AI-powered BMI-based personalized meal recommendations combined with seamless online food ordering.

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Introduction

Overview of the FitEats Project

FitEats is an intelligent, health-focused food delivery web application that integrates Artificial Intelligence with personalized nutrition recommendations. The platform addresses the critical gap in traditional food delivery systems — which prioritize convenience over health — by combining BMI-based dietary guidance with seamless online ordering.

The Problem

Most food delivery apps focus on speed and variety, ignoring nutritional value. Users order without knowing calorie content or suitability for their health condition.

The Solution

FitEats calculates the user's BMI using height & weight, categorizes them (underweight / normal / overweight / obese), and recommends personalized meals accordingly.

The Technology

Built on the MERN Stack — React.js frontend, Node.js + Express.js backend, MongoDB database — with JWT authentication and role-based access control.

Key Highlights

7+
System Modules

4
BMI Categories

MERN
Tech Stack

JWT
Auth Security

Motivation

Why FitEats? The driving forces behind the project

The rapid rise of online food delivery has created convenience — but at the cost of healthy eating. Rising obesity, lifestyle diseases, and lack of nutritional awareness demand a smarter solution.

Growing Health Awareness

Users increasingly seek apps that align with fitness goals and balanced nutrition.

Rising Lifestyle Diseases

Obesity, diabetes, and poor diet are major public health concerns linked to unguided food choices.

AI Advancements

Modern AI and recommendation systems make personalized nutrition guidance technically feasible.

Popularity of Food Apps

Massive growth in food delivery platforms creates an ideal channel for health-aware recommendations.

Nutritional Guidance Gap

No major food delivery platform bridges the gap between ordering convenience and health management.

Secure & Scalable Demand

Users expect trusted, secure applications with personalized yet private health data handling.

Problem Statement

Traditional food delivery applications focus on speed and variety — but ignore the most important factor: user health.

CORE PROBLEM

No existing platform combines food ordering with personalized BMI-based health recommendations

01
Unhealthy Ordering
Users order without nutritional awareness, consuming excess calories

02
No BMI Guidance
No app calculates BMI and adapts food suggestions to health status

03
Rising Obesity
Unguided food delivery contributes to lifestyle-related health issues

04
Generic Recommendations
Existing apps recommend based on popularity, not dietary needs

Therefore, there is a need for an intelligent, health-aware food delivery system integrating BMI analysis, AI recommendations, and convenient ordering.

Aim and Objectives

What FitEats sets out to achieve

AIM: To develop an intelligent, health-focused food delivery platform that combines BMI-based AI recommendations with seamless online ordering, promoting healthier food choices for users.

1. Develop a smart food delivery platform integrating AI-based meal recommendation
2. Calculate user BMI using height and weight to categorize health status
3. Recommend healthy meals based on BMI category and dietary preferences
4. Implement secure user authentication using JWT and role-based access control
5. Enable add-to-cart, order placement, and order management functionalities
6. Design a responsive and user-friendly web interface for all devices
7. Create an admin dashboard for managing food items, users, and orders
8. Improve user health awareness through personalized nutritional guidance

Literature Review

Existing research and published work in the domain

[2025] IJARCCE

FOOD DELIVERY WITH RECOMMENDATION SYSTEM

↳ Integrating recommendation logic with food delivery improves user satisfaction.

[2025] JTAECR — Shorbaji et al.

AI-Enabled Mobile Food-Ordering Apps

↳ AI-driven ordering boosts customer engagement and repeat usage.

[2025] K. Agrawal

AI in Personalized Nutrition & Food Recommendation

↳ Machine learning enables precise dietary guidance based on individual health metrics.

[2025] IJRASET — Bhuvaneshwari B.

Smart Food Ordering with Nutrition via Conversational Interface

↳ Chatbot-based interfaces improve nutrition guidance accessibility.

[2025] IJPREMS — Radadiya & Parekh

AI-Powered Food Delivery Web Application

↳ Full-stack AI food apps demonstrate practical viability for commercial deployment.

[2024] IJIRSET

AI Based Food and Diet Recommendation System

↳ BMI and preference data are effective inputs for recommendation engines.

Research Gap

Despite significant advances in AI-based food recommendation, critical gaps remain in existing literature and commercial systems:

⚠

No Real-time BMI Integration

Existing systems use static preferences. No platform dynamically calculates BMI and adapts recommendations in real time.

⚠

Lack of Health-Delivery Convergence

Food delivery apps and health apps remain separate. No platform merges both into one unified, intelligent solution.

⚠

No Nutritional Context in Ordering

Cart and order systems do not display calorie counts or nutritional suitability during the selection process.

⚠

Limited Medical Condition Support

Current research focuses on general BMI; conditions like diabetes, allergies, and hypertension are largely ignored.

⚠

Absence of Role-Based Health Admin

No system provides a health-aware admin dashboard for managing food items with nutritional metadata.

⚠

Mobile & Scalability Gap

Most research prototypes are not mobile-ready or scalable — limiting real-world deployment potential.

System Architecture

Three-tier MERN Stack architecture of FitEats

PRESENTATION TIER

- React.js Frontend
- Bootstrap 5 UI
- Responsive Design
- User Interface Pages

LOGIC / API TIER

- Node.js + Express.js
- REST API Endpoints
- JWT Authentication
- BMI Calculation Engine
- AI Recommendation Logic

DATA TIER

- MongoDB Database
- User Profiles & BMI
- Food Item Catalogue
- Order & Cart Records

DATA FLOW: User → React Frontend → REST API (Express) → Business Logic (BMI + AI Recommendation) → MongoDB → Response → User

Role-Based Access • JWT Token Security • Separate Admin & User Routes • Encrypted Credentials

Dataset & User Inputs

Data collected and used by the FitEats system

USER INPUT DATA	
Name & Email	Registration credentials
Password	Secured with encryption
Height (cm)	Used for BMI calculation
Weight (kg)	Used for BMI calculation
Food Preferences	Dietary choices & restrictions
Cart Selections	Food items added to cart
Order Details	Confirmed purchases

FOOD & SYSTEM DATASET	
Food Name	Item label for display
Category	Veg / Non-Veg / Vegan
Calories (kcal)	Caloric content per serving
Nutritional Tags	High-protein, low-carb, etc.
BMI Suitability	Underweight / Normal / Overweight / Obese
Price (₹)	Cost per item
Admin-Added	Managed via admin dashboard

Proposed Methodology

Agile development approach with modular phases

01 **Requirement Analysis**

Identified functional & non-functional requirements for users, admins, and the recommendation engine.

02 **System Architecture Design**

Designed three-tier MERN architecture with REST APIs, JWT auth, and MongoDB schema.

03 **BMI Calculation Logic**

Implemented BMI = Weight(kg)/Height(m)^2 to classify users into 4 health categories.

04 **Food Recommendation Engine**

Mapped BMI categories to curated food lists filtered by calorie range and nutritional value.

05 **Authentication & Security**

JWT-based login, bcrypt password hashing, role-based access for user and admin routes.

06 **Cart & Order Management**

Cart module with quantity tracking, price calculation, and order confirmation workflow.

07 **Testing & Validation**

Unit, integration, system, and user acceptance testing across all modules.

BMI Recommendation System

The core intelligence behind FitEats

BMI Formula: BMI = Weight (kg) ÷ Height² (m²)

$$BMI = Weight (kg) \div Height^2 (m^2)$$

< 18.5 Underweight	18.5 - 24.9 Normal Weight	25 - 29.9 Overweight	≥ 30 Obese
Recommended Foods:	**Recommended Foods:**	**Recommended Foods:**	**Recommended Foods:**
High-calorie dense meals	Balanced diet meals	Low-calorie options	Very low calorie meals
Protein-rich foods	Mixed macro nutrition	High-fiber foods	High-protein, low-fat
Nuts, seeds, dairy	Variety of food groups	Lean proteins	Steamed/grilled food
Complex carbohydrates	Maintain current intake	Reduced carbohydrates	No sugary/fried items

Modules of the System

Seven functional modules powering FitEats

User Authentication

Secure JWT-based login, registration, bcrypt password hashing, session management, and role-based access.

BMI Calculation

Calculates BMI from user height/weight, categorizes into 4 health groups, stores records in database.

Food Recommendation

AI-logic maps BMI category to food items filtered by calorie range, nutritional tags, and preferences.

Add to Cart

Users add, update, and remove food items. Calculates item price, quantity, delivery charge, and total.

Food Ordering

Complete order placement flow: item selection → cart → confirmation → order storage in database.

Database Management

MongoDB stores users, BMI records, food catalogue, cart data, orders, and admin configurations.

Admin Dashboard

Admins can add/update/delete food items, view all user orders, and manage the complete food catalogue.

Technologies Used

FRONTEND	BACKEND	DATABASE
React.js SPA framework	**Node.js** Runtime environment	**MongoDB** NoSQL database
HTML5 & CSS3 Structure & style	**Express.js** REST API framework	**Mongoose ODM** Schema modeling
Bootstrap 5 Responsive UI	**JWT** Authentication tokens	**Atlas / Local** DB hosting
JavaScript Interactivity	**bcrypt** Password hashing	**Indexes** Query optimization

Development Tools: VS Code • Postman • Git & GitHub • npm • MongoDB Compass • Browser DevTools

Implementation

Development environment and experimental setup

EXPERIMENTAL SETUP	
Processor	Intel Core i5 / i7
RAM	8 GB or above
OS	Windows 10 / 11
Frontend	React.js + Bootstrap 5
Backend	Node.js + Express.js
Database	MongoDB
Auth	JWT + bcrypt
IDE	Visual Studio Code

IMPLEMENTATION PHASES

- Frontend UI Development
- Backend API Development
- Database Schema Design
- BMI Engine Integration
- Recommendation Logic
- Authentication Setup
- Cart & Order System
- Admin Dashboard
- Integration Testing

User Interface Design

Key screens of the FitEats application

🏠 Home Page

Welcome screen with featured foods, navigation menu, and call-to-action for BMI check.

👤 Register / Login

Secure sign-up and sign-in form with validation, error messages, and session tokens.

⚖️ BMI Calculator

Height & weight input form, real-time BMI result display with health category badge.

Recommendation Page

Personalized food grid filtered by BMI category — shows calories, nutrition tags, and price.

🛒 Cart & Checkout

Item list with quantity controls, subtotal, delivery charges, and order confirmation button.

⚙️ Admin Dashboard

Food management panel: add/edit/delete items, view orders, manage users and catalogue.

Results and Discussion

Experimental outcomes and performance evaluation

User Authentication	✓ Successful	Secure registration, login, session management, and JWT token handling worked without failures across all test cases.
BMI Calculation	✓ Accurate	BMI correctly calculated for all input ranges. Categories (Underweight/Normal/Overweight/Obese) assigned accurately.
Food Recommendation	✓ Personalized	Correct food items served per BMI category. Calorie and nutritional filters applied successfully.
Cart Management	✓ Functional	Item add/remove, quantity update, price calculation (subtotal + delivery) computed correctly.
Order Processing	✓ End-to-End	Full ordering flow tested: item → cart → confirm → database storage → order confirmation displayed.
Admin Dashboard	✓ Operational	Admin can add, update, and delete food items. Order list view and user management functional.

Comparative Analysis

FitEats vs. Traditional Food Delivery Platforms

Feature	Swiggy / Zomato	Generic Apps	FitEats ✓
BMI Calculation	✗	✗	✓
Personalized Recommendations	Partial	✗	✓ Full
Calorie-Based Filtering	✗	✗	✓
Health Category Assignment	✗	✗	✓
Nutritional Guidance	✗	✗	✓
JWT Authentication	✓	Partial	✓
Admin Food Management	✓	Partial	✓
BMI-linked Food Catalogue	✗	✗	✓
Role-Based Access Control	✓	Partial	✓
Health-Focused Ordering	✗	✗	✓

Work Completed

Summary of all completed project deliverables

Phase 1: Planning

- Problem identification and research
- Literature review across 10+ papers
- Requirement analysis and scope definition

Phase 2: Design

- System architecture design (MERN)
- Database schema and ER diagram
- UI wireframes and flow diagrams

Phase 3: Development

- React.js frontend with Bootstrap
- Node.js + Express.js REST APIs
- MongoDB integration via Mongoose

Phase 4: Core Features

- BMI calculation and classification
- AI-based food recommendation engine
- JWT authentication and security

Phase 5: Modules

- Cart and order management system
- Admin dashboard for food management
- User profile and session handling

Phase 6: Testing

- Unit, integration, and system testing
- User acceptance testing (UAT)
- Bug fixes and performance optimization

Current Limitations

Known constraints of the present system version

⚠ Limited Medical Analysis

The recommendation engine relies only on BMI. Advanced conditions like diabetes, hypertension, allergies, and chronic diseases are not considered.

⚠ No Real-Time Delivery Tracking

The system does not include live GPS-based delivery tracking or estimated delivery time features.

⚠ No Online Payment Gateway

Financial transactions are not yet integrated. The system does not support UPI, cards, wallets, or net banking.

⚠ Web-Only Platform

FitEats is currently a web application only. Dedicated Android or iOS mobile apps are not yet developed.

⚠ Basic Recommendation Logic

Food recommendations are based on rule-based BMI mapping rather than advanced ML models trained on user behavior.

⚠ No AI Chatbot Support

There is no conversational AI interface for nutrition guidance or customer support integrated in the current version.

Future Scope

Planned enhancements to make FitEats commercially viable and more intelligent

Advanced AI & ML

Deep learning models, neural networks, and user behavior analysis for smarter recommendations.

Real-Time GPS Tracking

Live delivery monitoring, route optimization, and estimated arrival time.

Payment Gateway

UPI, credit/debit cards, net banking, QR code, and digital wallet integration.

Mobile Apps

Android and iOS apps using React Native or Flutter for wider accessibility.

Medical Diet Plans

Diet recommendations for diabetes, hypertension, allergies, and specific medical conditions.

Voice Ordering

Voice-based food ordering system for hands-free, accessible user interaction.

Cloud Deployment

AWS / Azure / GCP deployment for large-scale availability and elastic scalability.

AI Chatbot

Conversational AI for nutrition guidance, order tracking, and customer support.

Conclusion

FitEats: Intelligent Food Delivery System successfully achieves its core objective — bridging the gap between online food delivery and personal health management.

- Developed a complete MERN-stack intelligent food delivery platform
- Integrated real-time BMI calculation with health category classification
- Implemented AI-based food recommendation filtered by nutritional criteria
- Secured the system with JWT authentication and role-based access control
- Built full cart, order management, and admin dashboard functionalities
- Validated all modules through unit, integration, and system testing
- Demonstrated practical AI application in the nutrition and delivery domain

FitEats proves that AI and health awareness can coexist with the convenience of modern food delivery systems.

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